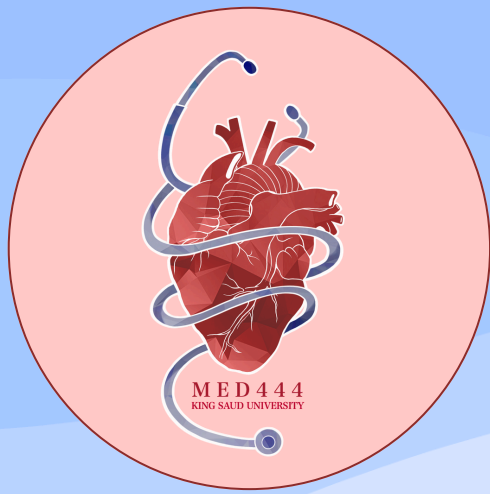


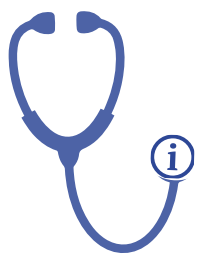


INTRODUCTION TO MEDICAL INFORMATICS

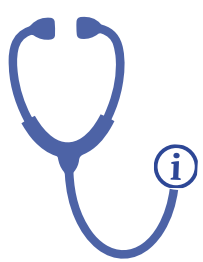
Editing files

- main text- male slides
- female slides- important
- dr notes - textbook
- extra / old slides

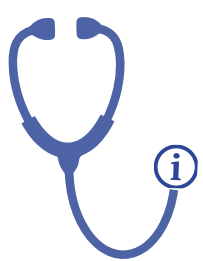
objectives



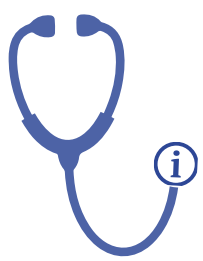
Describe the concept of medical and Biomedical informatics



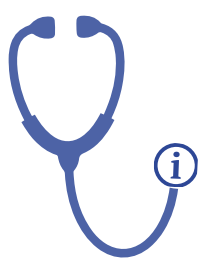
Describe the concepts of evidence-based medicine



List the key components and underlying concepts of medical informatics



Identify the main benefits and challenges of medical informatics



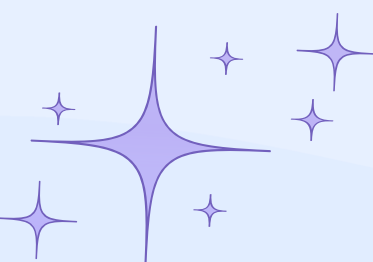
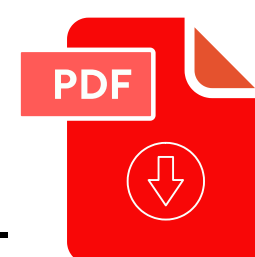
Describe the concepts of telemedicine, telehealth and e-health



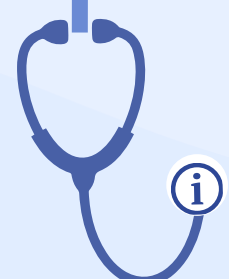
Describe data, information, and knowledge

**The lecture was presented through BlackBoard
By: Prof. Ahmed Albarrak**

**[Click here to view the resource that was
attached with the slides in the Blackboard](#)**



اللهم انفعني بما علّمتني، وعلمني ما ينفعني، وزدني علماً





Introducton to Medical Informatics

What is informatics?

The **science** concerned with acquisition,gathering manipulating,storing,retrieving and classifying recorded information.

Greenes & Shortliffe 1990

What is Medical Informatics?

the field of the cognitive information processing, and communication tasks in medical practice, education, and research, including the information science and the technology to support these tasks.

1

Medical informatics is a rapidly developing scientific field that deals with the storage, retrieval, and optimal use of biomedical information, data, and knowledge for problem solving and decision making.

Blois, M.S., and E.H. Shortliffe. in Medical Informatics: Computer Applications in Health Care, 1990, p. 20.

2

- Medical informatics is a developing body of knowledge and a set of techniques concerning the organizational management of information in support of medical research education, and patient care

3

Medical informatics combines medical science with several technologies and disciplines in the information and computer sciences and provides methodologies by which these can contribute to better use of the medical **knowledge** base and ultimately to better medical care.

in this definition they included data & information with knowledge

definition by Association of American Medical Colleges

4

Medical informatics attempts to provide the theoretical and scientific basis for the application of computer and automated information systems to biomedicine and health affairs
medical informatics studies biomedical information, data, and knowledge their storage, retrieval, and optimal use for problem-solving and decision-making

Lindberg, D.A.B. NLM Long Range Plan. Report of the Board of Regents, 1987, p. 31.

5

Medical informatics is the application of computers, communications and information technology and systems to all fields of medicine - medical care, medical education and medical research.

definition by MF Collen (MEDINFO '80, Tokyo, later extended).

6

Medical informatics comprises the theoretical and practical aspects of information processing and communication, based on knowledge and experience derived from processes in medical and Healthcare.

Van Bommel, J.H. "The structure of medical informatics" Medical Informatics, 9(1984), p. 175.

7

Medical Informatics is the interdisciplinary study of the design, development, adoption and application of IT-based innovations in healthcare services delivery, management and planning."

<https://www.himss.org/clinical-informatics/medical-informaticsHIMSSaccessed%20Feb%202019>

Origin of term Medical Informatics

Russian

informatika **1968** by AI Mikhailov, "Oznoy Informatika" ("Foundation of Informatics") structure and properties of scientific information

French

informatique de medecine
1968 university departments established with this title

English

first appeared in **1970s**
Columbia University changed its name from Medical Information Science to medical informatics

History Of Medical Informatics



History Of Medical Informatics

Early names of medical informatics included medical computing, medical computer science, computer medicine, medical electronic data processing, medical automatic data processing, medical information processing, medical information science, medical software engineering and medical computer technology.

In France in 1968 university departments established with the title "informatique de medecine"
In 1968 homer Warner, founded the department of medical informatics at the university of Utar.

1968

In the United States in 1996 , HIPAA (Health Insurance Portability and Accountability Act) regulations concerning privacy and medical record transmission created the impetus for large numbers of physicians towards using EMR (Electronic Medical Record) to move software, primarily for the purpose of secure medical billing.

1996

March 2001, CROSSING THE QUALITY CHASM: A NEW HEALTH SYSTEM FOR THE 21ST CENTURY

The report found that most errors described in the literature were linked to poor or unclear data quality. To address this, they recommended making Computerized Physician Order Entry (CPOE) a key priority. Since CPOE requires strong infrastructure (especially an Electronic Health Record system), they emphasized that it should be treated as a quality and safety project, not just as automation.

2001

HIPAA in US was the first act in the world that focused on the privacy of the patient record

1950s

Medical informatics as a science history started in the 1950s with the growth of devices, and computer applications in medicine.
The earliest reported use of computation for medicine was in dental projects in the 1950's at the National Bureau of Standards by Robert Ledley.
In the mid 1950s was the development of expert systems (decision support systems) such as MYCIN (first medical informatics system) and INTEREST-I.

at first They were focusing on automation but become more focused on data processing

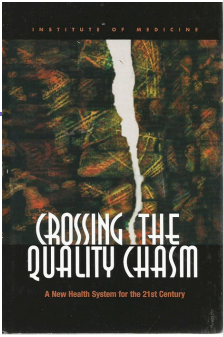
1970

In 1970 the international medical informatics association was founded. In the same year the MUMPS language and operating system was developed and used for clinical applications

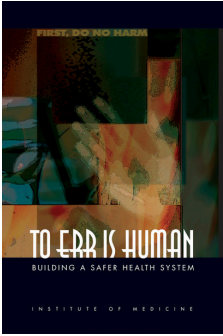
1999

Nov 1999, TO ERR IS HUMAN: BUILDING A SAFER HEALTH SYSTEM, by the institute of Medicine

(438) Analysis of previous report (to err is human) found that more than 50% of medical errors due to mishandling with patient information.
- (438) High recommendation to implement EMR for safety of patient not for automation.



- (438) Based on retrospective studies in new york & pittsburg, that alarmed the medical community about increased mortalities due to medical errors.



TEXTBOOK: Health information technology (HIT) are single-entry points into a clinical world in which computational tools assist not only with patient-care matters (reporting results of tests, allowing direct entry of orders or patient information by clinicians, facilitating access to transcribed reports, and in some cases supporting telemedicine applications or decision-support functions) but also administrative and financial topics , research ,scholarly information, even office automation

Health Informatics

Health Informatics is the intersection of information science , Computer sciences or Information Technology , and health care

1

It deals with resources, devices, & methods required to optimize the acquisition, storage, retrieval, and use of information in health care

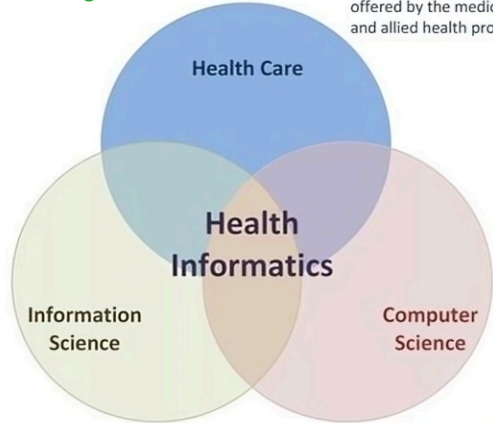
2

Health informatics tools include **clinical guidelines** , formal medical terminologies, information & **communication systems**. It is applied to the areas of nursing,clinical care, dentistry, pharmacy, public health and medical research.

medical informatics focus on the **communication systems** unlike the informatics which focus only on storage, retrieval

The area where we practice for treatment, prevention and management of disease

The prevention, treatment and management of illness and the promotion of health and well-being through the services offered by the medical, nursing and allied health professions



focusing on the tools, applications that will be used for this data

The collection, classification, manipulation, storage, retrieval and dissemination of information;
The application and use of knowledge;
The interaction between people, organisations and information systems.

The theoretical foundations of information and computation together with their implementation and application in computer systems

442 NOTES: The difference between information, data and knowledge: A patient's blood sugar is 140 --> data. A patient's blood sugar is 140 and you know that the normal range of blood sugar is less than 125 = patient has hyperglycemia --> Information
Another example Student exam The Grades= Data The mean and median of the grades to the student are information If we compare between the new and old grades to know if they increase or decrease=knowledge . Compare the results with research or data -> knowledge.

438 notes: -Data: raw material, not processed and it is not only number; it can be graphs, Data alone does not provide any meaning unless the person has a reference and a knowledge about this data.
-Information: analysis. You start to refer the data to another value to have a meaning that can be understood and later be compared to other information to get the knowledge. Examples: are mean and standard deviation.
-Knowledge: interpretation, you are trying to get a message from the information you have; you refer to other information and compare.
-In a research, data is what you collect, your target population.
Information is the result of the analysis of data. Knowledge is the discussion, conclusion and recommendations.

Biomedical Informatics



Biomedical Informatics

Biomedical informatics is the bigger umbrella for health informatics.

- Biomedical Informatics (BMI): is the **interdisciplinary** field that studies and pursues the effective uses of biomedical data , information, and knowledge for **scientific inquiry** (ex: using patients' medical records for research), problem solving (ex: telemedicine to reach patients) , and decision making (ex: drug library), driven by efforts to improve human health

E.H. Shortliffe and Marsden S. Blois 2014

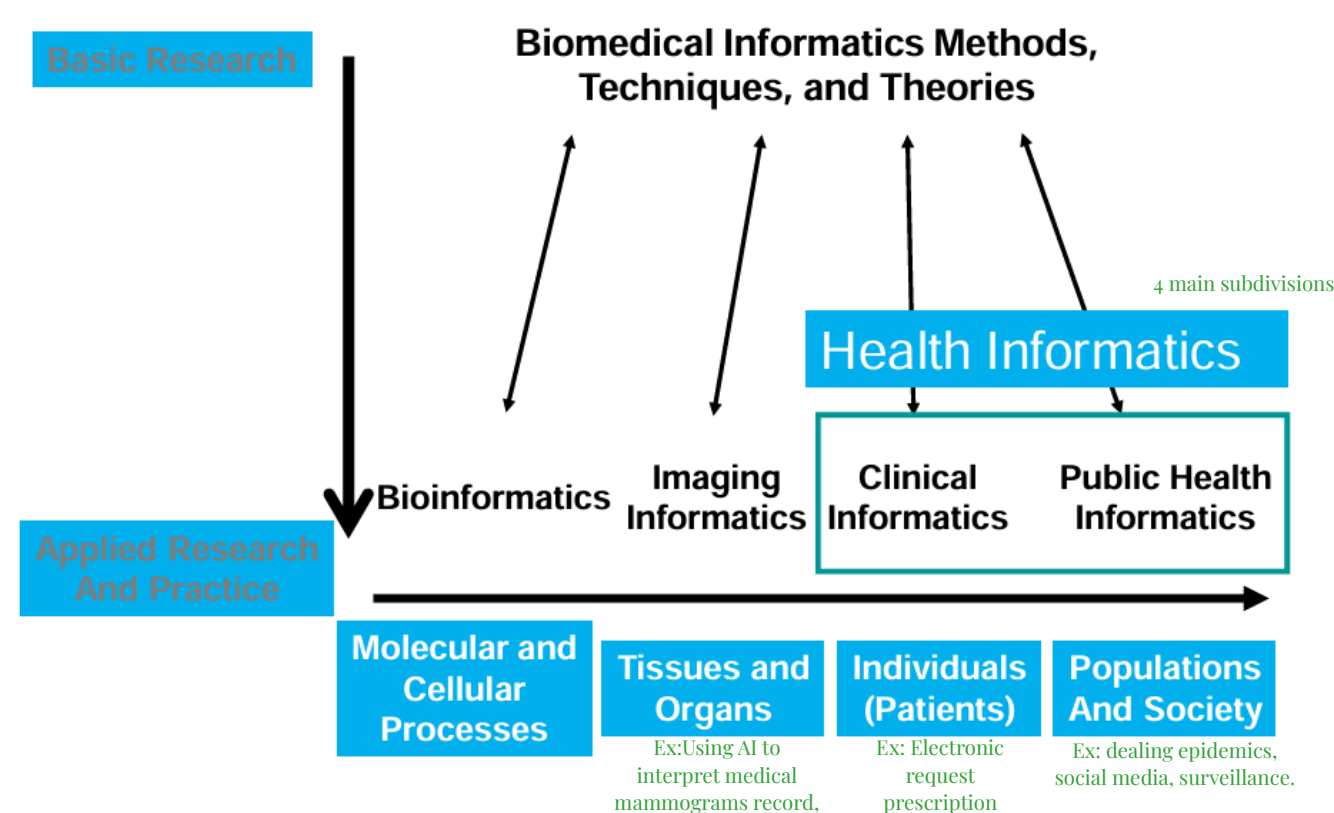
Scope and breadth of the discipline:

BMI investigates and supports reasoning, modeling(نمذجة), simulation, experimentation, and translation across the spectrum from molecules to individuals and to populations, from biological to social systems, bridging basic and clinical **research** and **practice** and the health care enterprise.

It means that Biomedical informatics is connecting this spectrum from the molecular level to the population + biological and social system to clinical both in research and practice

Biomedical Informatics in Perspective

The tools of biomedical informatics support the basic research and link it with the applied research



TEXTBOOK: - Biomedical informatics is, by its nature, an experimental science, characterized by posing questions, designing experiments, performing analyses, and using the information gained to design new experiments. One goal is simply to search for new knowledge, called basic research. A second goal is to use this knowledge for practical ends, called applications (applied) research. - biomedical informatics (BMI) is inherently motivated by problems encountered in a set of applied domains in biomedicine. The first of these historically has been clinical care (including medicine, nursing, dentistry, and veterinary care), an area of activity that demands patient-oriented informatics applications. We refer to this area as clinical informatics. It includes several sub-topics and areas of specialized expertise, including patient-care foci such as nursing informatics, dental informatics, and even veterinary informatics. Furthermore, medical informatics reserved for those applied research and practice topics that focus on disease and the role of physicians. Closely tied to clinical informatics is public health informatics. Two other large areas of application overlap in some ways with clinical informatics and public health informatics. These include imaging informatics (and the set of issues developed around both radiology and other image management and image analysis. Finally, there is the burgeoning area of bioinformatics, which at the molecular and cellular levels is offering challenges that draw on many of the same informatics methods as well. At the next level, workers focus on tissues and organs, which tend to be the emphasis of imaging informatics work (also called structural informatics).

- Closely tied to clinical informatics is public health informatics. where similar methods are generalized for application to populations of patients rather than to single individuals. clinical informatics and public health informatics share many of the same methods and techniques.

Theory and methodology

BMI develops, studies, and applies theories, methods, and processes for the generation, storage, retrieval, use, management, and sharing of biomedical data, information, and knowledge.

Technological approach

BMI builds on and contributes to computer, telecommunication, and information sciences and technologies, emphasizing their application in biomedicine.

Application areas

range from bioinformatics to clinical and public health informatics and span the spectrum from the molecular to population levels of health and biomedicine. "

AMIA amiajnl-2012

AMIA amiajnl-2012-001053 1..8

The Fundamental Theorem of Biomedical Informatics:

Charles friedman proposed the fundamental theorem of biomedical informatics which states that "a person working with an information resource is better than that same person unassisted".

Biomedical informatics help the human to make a decision and solve a problem , but we shouldn't compare human with machine, while the machine have very fast speed in processing information it doesn't have the cognitive function to make a decision or recognize something the machine not trained on



Bioinformatics:

Bioinformatics help researchers when they want to find the relation of a certain gene causing a certain disease and linking it with the symptoms

- The collection, organization, and analysis of large amounts of biological data, using computers and databases.
- Historically, bioinformatics concerned itself with the analysis of the sequences of genes and their products (proteins), but the field has since expanded to the management, processing, analysis, and visualization of large quantities of data from genomics, proteomics, drug screening, and medicinal chemistry.
- Bioinformatics also includes the integration and "mining" of the ever-expanding databases of information from these (biological) disciplines.

Applications of Medical Informatics



1

Telemedicine:

is composed of the Greek word $\tau\epsilon\lambda\epsilon$ (tele) meaning 'far', and medicine. It is therefore the delivery of medicine at a distance. A more extensive definition is that it is the use of telecommunication and information technologies for the provision of clinical care to individuals located at a distance and to the transmission of information to provide that care.

- (438) Examples of telemedicine:

1. Teleconsultation (this requires the consultant to have an access to the patient information so he/she can do the consultation).
2. Teleradiology.

2

Telehealth:

The delivery of health related services, enabled by the innovative use of technology, such as videoconferencing, without the need for travel.

-(438) Telehealth was established because of the aging age group who need a support especially those who are lonely to provide support at home and also for home monitoring and health education

telemedicine	telehealth
clinical services (doc-patient)	related health services (everything else)

3

Hospital Information System:

HIS: is a comprehensive information system dealing with all aspects of information processing in a hospital.

- This encompasses human (and paper-based) information processing as well as data processing machines.
- As an area of Medical Informatics the aim of HIS is to achieve the best possible support of patient care and administration by electronic data processing.

HIS elements:

1. Patients care (clinical documentation, order entry, OR...)
2. Patients administration (registration, outpatient, inpatient, ER...)
3. Financial & admin services.
4. Materials (inventory, purchase, engineering management...)
5. Clinical support (labs, pharmacy, radiology...)





4

Ehealth:

Also written e-health, is a relatively recent term for healthcare practice which is supported by electronic processes and communication, some would argue the term is interchangeable with Health Informatics.

Tele-Health or e-health E-health:

is much more than tele-health as tele is a limiting factor to the form of technology in health. E-health could be at distance or local.

E- Health: هم تحت E-Health, Telemedicine e prescription, أي استخدام إلكتروني لتطبيقات صحية

From Telemedicine to eHealth:

- The practice of telemedicine will become more prominent and will be part of the mainstream of Healthcare;
- It will become increasingly difficult to differentiate telemedicine from many other uses of technology in the delivery of healthcare;
- E-health is all inclusive and captures the use of Internet technologies and the rise of the information economy. This includes: – information technology; – telecommunication technology; – Data transmission protocols and techniques;
- E-Health is all inclusive and captures all types of Healthcare and Healthcare professionals: it is not limited to medicine and not limited to doctors;

E-Health components:

Four essential components make the e-health:

- Medical knowledge (data, information, knowledge) that lends itself to being stored in computer files **(digital format)** If they were not in digital format they will be difficult to use and analyze them using the health informatics methods
 - People who are willing/able to share, apply and use this knowledge
 - Data processing equipment to record, store and process this data التطبيقات والأجهزة التي نستخدمها لحفظ هذي المعلومات واسترجاعها وتحليلها
 - Telecommunication facilities to transfer (exchange) this data electronically between remote locations. البنية التحتية من الشبكات والاتصالات سواء بنفس المستشفى أو خارج المستشفى
- (438) E-health was used for business reasons (selling and buying medical devices and drugs). Nowadays is used for all applications used for healthcare such as heartbeat monitors .. etc

5

Electronic medical record (EMR):

EMR is A general term describing computer-based patient record systems. It is sometimes extended to include other functions like order entry for medications and tests, amongst other common functions.

- EMR is important for decision support.
- it's not just reflection of paper medical records, it's more structured and organized and it has a decision support system
- if a patient had a pacemaker and you wanted to do an MRI, this system will prevent the MRI from taking place avoided the complications.
- Another example is drug-drug interaction and allergies

Applications of Medical Informatics



6

Evidence Based Medicine:

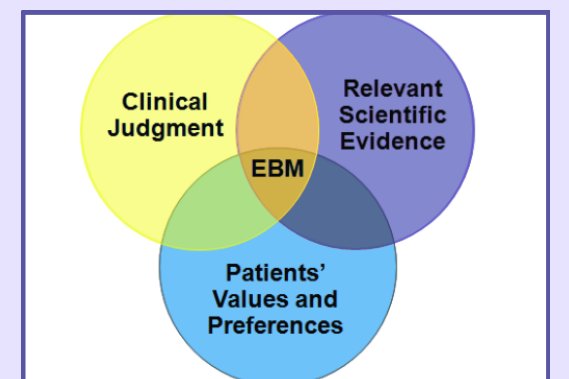
Entails a system that provides information on appropriate treatment under certain patient conditions. A healthcare professional can look up whether his/her diagnosis is in line with scientific research. The advantage is that the data can be kept **up-to-date** with published knowledge

“is the process of systematically reviewing, appraising and using clinical research findings to aid the delivery of optimum clinical care to patients.”

'the integration of best research evidence with clinical expertise and patient values' which when applied by practitioners will ultimately lead to improved patient outcome. EBM falls under the education area in health informatics

There are three main pillars or components of EBM:

1. Best evidence (**Relevant Scientific Evidence**): clinical research that has been conducted using rigorous methodology. **mainly randomized clinical trials and systematic review**
2. Clinical expertise (judgment) refers to the clinician's cumulated education, experience and clinical skills.
3. Patients preferences, values and concerns which patient brings to a clinical encounter.



It is clear that EBM depends on the structured knowledge databases that contains the most recent and valid clinical research output. It is the integration of best and latest evidence with clinical expertise and patient values component could improve the clinical practice outcome and patients' satisfactions.

- Why EBM is part of informatics? Because all knowledge must be up to date and get these we need a medium for that in which all can reach and also a fast one.
- What is the difference between evidence based practice and none? Latest knowledge!
- EBM is found to reduce errors. The knowledge has to be structured.



7

Dental informatics:

Is the name given to the application of information technology to dentistry. It is often considered a subset of Medical Informatics and Biomedical Informatics.

- (438) Dental practice use medical informatics more than medical practice because these practitioners take the case from A to Z alone.

8

Distance Learning:

- With aid of telecommunications technologies and internet, distance learning is now widely applied in many universities, eg Open University
- It is now possible to earn university degrees from home, at every level from bachelor's to doctorate.

Applications of Medical Informatics



9

Nursing informatics :

Nursing Informatics is a specialty of Health Informatics (like Medical Informatics, Consumer Health Informatics, and Telehealth) which deals with the support of nursing by information systems in **delivery**, documentation, administration and evaluation of patient care and prevention of diseases.

438 notes:

- Nursing depend on informatics a lot because they do a lot of documentations.
- Another example, a nurse can't give a patient a medication until they check the patient code with a device that has all patients information about the drugs.



10

Continuing Medical Education (CME):

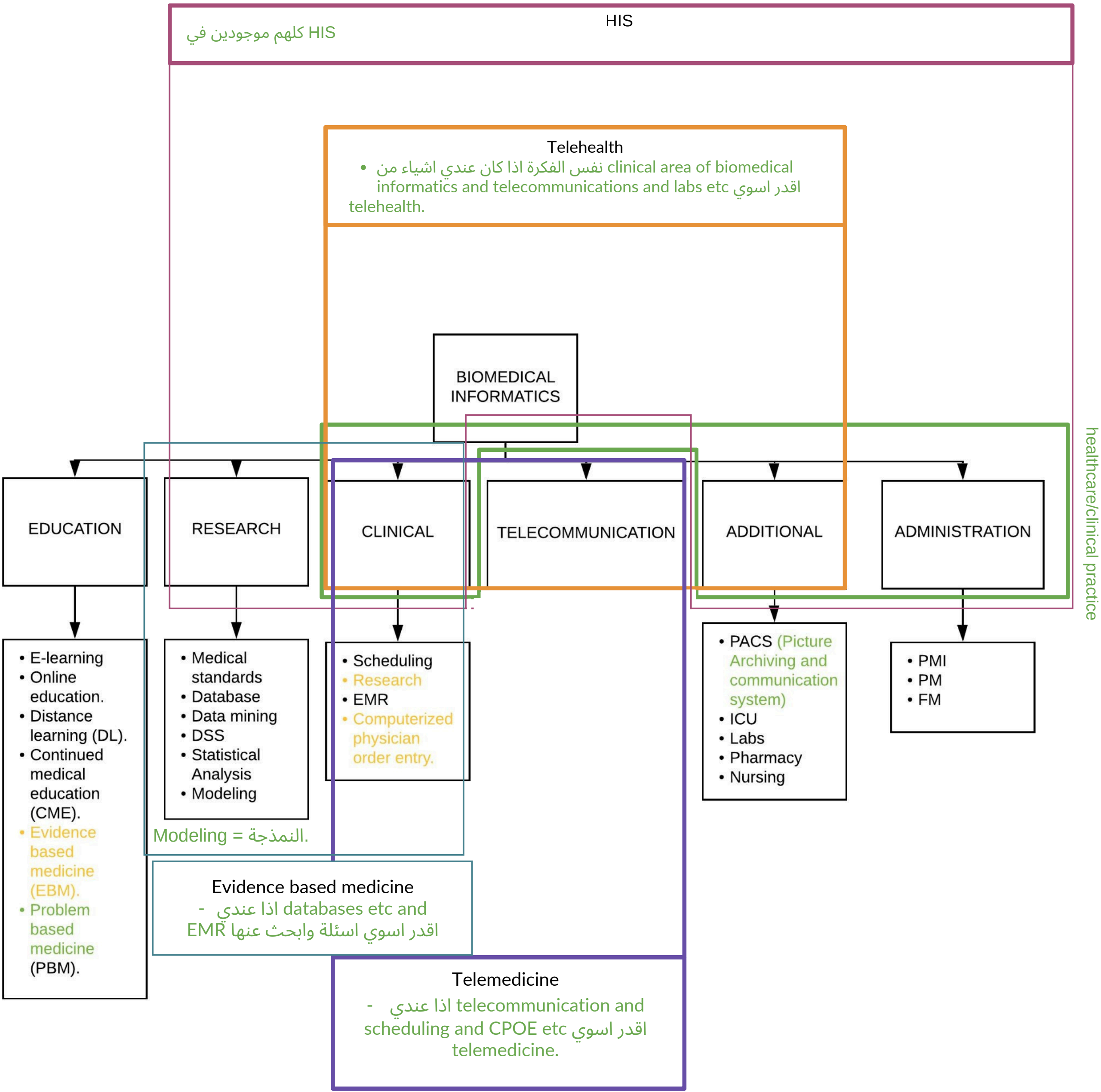
- The science of medicine advances at such a rapid rate that much of what is taught becomes outmoded, and it has become obligatory for physicians to be lifelong learners, both for their own satisfaction and, increasingly, as a formal government requirement to maintain licensure.
- Doctors who practice in rural areas or other more isolated locations may face considerable obstacles to obtain hours for CME.
- The cost of web-based or **online** CME is much lower than the cost of traditional CME.
- (438) Blackboard used in our college is an example and also DXR.
- help to deliver information for physicians in remote areas

WHY MEDICAL INFORMATICS for healthcare?

- | | |
|--|---|
| <div>1</div> <div>Improve Healthcare quality</div> <div>& Helps to establish guidelines</div> | <div>9</div> <div>Reduce duplication of efforts</div> <div>such as :doing a test more than one time because it wasn't recorded the first time</div> |
| <div>2</div> <div>Better data access</div> <div>Especially with EMR where we have clear data with proper terminology and no hand writing</div> | <div>10</div> <div>Provide unified access to all existing data</div> |
| <div>3</div> <div>Faster data retrieval and storage</div> <div>Such as : Allergies - Lab results</div> | <div>11</div> <div>Increase healthcare organization efficiency</div> |
| <div>4</div> <div>High quality data</div> <div>No missing values</div> | <div>12</div> <div>Reducing cost and achieves quality of healthcare</div> |
| <div>5</div> <div>Support medical and non-medical decision-making</div> <div>Such as : Accessing to see the DDI when ordering a medication
non medical such as : manpower - workflow</div> | <div>13</div> <div>Improve staff productivity</div> <div>And cooperation</div> |
| <div>6</div> <div>Enhance quality assurance</div> <div>Monitor performance: ER waiting time , how many beds available in ER , what is the LOS of a pt after surgery</div> | <div>14</div> <div>Reduce redundant tests, services and information entry</div> |
| <div>7</div> <div>Enhance out-come researches and studying programs</div> | <div>15</div> <div>Manage billing and payment system</div> |
| <div>8</div> <div>Sharing medical data</div> <div>Either within the same hospital such as : from radiology department to general surgery or with other hospitals</div> | <div>16</div> <div>Eliminate and reduce errors</div> |

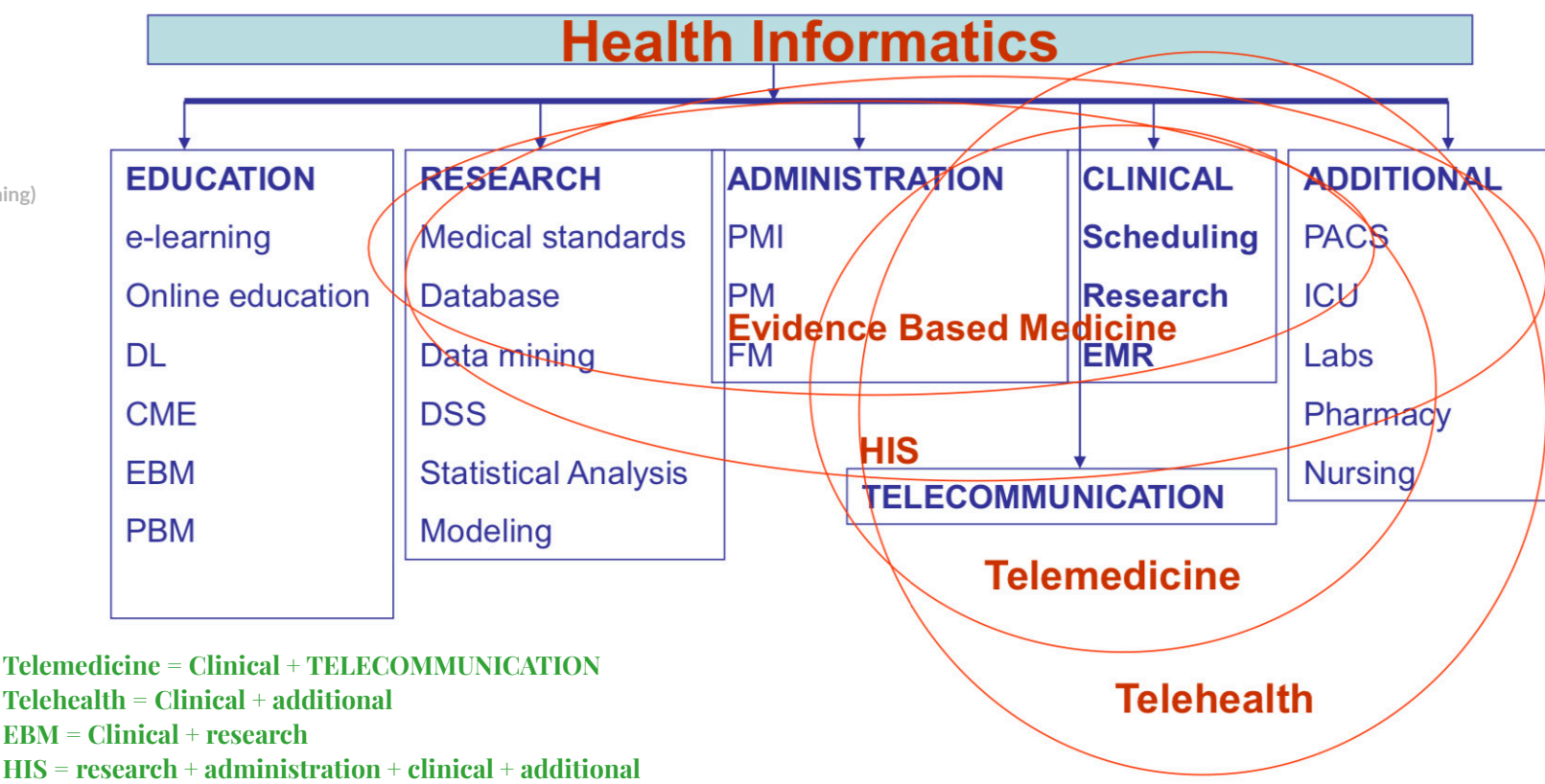


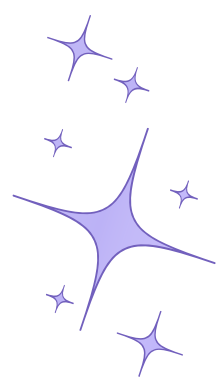
All the notes in this page are from 438's R.



Original diagram for doctor's slides

- DL → Distance Learning
- CME → Continuing Medical Education
- EBM → Evidence-Based Medicine
- PBM → Problem-Based Medicine (or Problem-Based Learning)
- DSS → Decision Support Systems
- PMI → Patient Medical Information
- PM → Practice Management
- FM → Financial Management
- EMR → Electronic Medical Record
- PACS → Picture Archiving and Communication System
- ICU → Intensive Care Unit
- HIS → Hospital Information System





SUMMARY



Informatics	The science concerned with gathering, manipulating, storing, retrieving and classifying recorded information.
Medical Informatics	The field that concerns itself with the cognitive, information processing, and communication tasks of medical practice, education, and research, including the information science and the technology to support these tasks. - is a rapidly developing scientific field that deals with the storage, retrieval, and optimal use of biomedical information, data, and knowledge for problem solving and decision making. - The difference between information and data: A patient's blood sugar is 140 --> data. A patient's blood sugar is 140 and you know that the normal range of blood sugar is less than 125 = patient has hyperglycemia --> Information. - Medical informatics comprises the theoretical and practical aspects of information processing and communication, based on knowledge and experience derived from processes in medical and Healthcare. - The next step in the mid 1950s was the development of expert systems (decision support systems) such as MYCIN (first medical informatics system) and INTEREST-I. - Informatika = one of the origin terms of medical informatics used by Russia. - In the United States in 1996, HIPAA (Health Insurance Portability and Accountability Act) concerning privacy and medical record transmission created the impetus for large numbers of physicians to move towards using EMR (Electronic Medical Record) software, primarily for the purpose of secure medical billing. - Better data access and Enhance quality assurance are advantages of the use of medical informatics in health care.
Health Informatics	- Is the intersection of information science, Information Technology, and health care. Computer science = information technology - Include clinical guidelines, formal medical terminologies, information & communication systems. is applied to the areas of nursing, clinical care, dentistry, pharmacy, public health and (bio)medical research.
Biomedical Informatics (BMI)	Is the interdisciplinary field that studies and pursues the effective uses of biomedical data , information, and knowledge for scientific inquiry (ex: using patients' medical records for research) , problem solving (ex: telemedicine to reach patients) , and decision making (ex: drug library), driven by efforts to improve human health. - Biomedical informatics is the bigger umbrella for health informatics. - Application areas: Range from bioinformatics to clinical and public health informatics and span the spectrum from the molecular to population levels of health and biomedicine. “ - Health informatics subfields are clinical informatics and public health informatics. - Ex of clinical informatics is EMR. - Public health informatics is concerned with clinical application on the population.
Bioinformatics	- The collection, organization, and analysis of large amounts of biological data, using computers and databases. - Historically, bioinformatics concerned itself with the analysis of the sequences of genes and their products (proteins), but the field has since expanded to the management, processing, analysis, and visualization of large quantities of data from genomics, proteomics, drug screening, and medicinal chemistry. - Includes the integration and “mining” of the ever-expanding databases of information from these (biological) disciplines.
Telemedicine	2 Definitions: - Delivery of medicine at a distance. - The use of modern telecommunication and information technologies for the provision of clinical care to individuals located at a distance and to the transmission of information to provide that care.
Telehealth	The delivery of health related services, enabled by the innovative use of technology, such as videoconferencing, which the need of travel.
E-health	healthcare practice which is supported by electronic processes and communication, some people would argue the term is interchangeable with Health Informatics. - Four essential components make the e-health: - Medical knowledge (data, information, knowledge). - People who are willing/able to share, apply and use this knowledge. - Data processing equipment to record, store and process this data. - Telecommunication facilities to transfer (exchange) this data electronically between remote locations.
Evidence based medicine	- Is the process of systematically reviewing, appraising and using clinical research findings to aid the delivery of optimum clinical care to patients.” - A healthcare professional can look up whether his/her diagnosis is in line with latest (up-to-date) scientific research. - The integration of best research evidence with clinical expertise and patient values. - It depends on the structured knowledge databases that contains the most recent and valid clinical research output.



1- is the application of computers, communications and information . technology and systems to all fields of medicine : medical care, medical education and medical research

A) Medical research

B) Biomedical Informatics (BMI)

C) Medical Informatics

2-Which one of the following is not a component of the health informatics Triad

A) information science

B) BioInformatics

C) Information Technology

3-he interdisciplinary field that studies and pursues the effective uses of information, knowledge for scientific inquiry problem solving , and decision making driven by efforts to improve human health

A) Telemedicine

B) BioInformatics

C) Biomedical Informatics

4- Which of the following best differentiates eHealth from Telehealth?

A) Telehealth is broader and includes all healthcare services using Technology, while eHealth is limited to distance care

B) eHealth is broader and includes both local and distance electronic healthcare, while Telehealth is limited to distance care

C) Telehealth involves electronic record-keeping, while eHealth is only about patient communication

5-The three main pillars of Evidence-Based Medicine (EBM) are:

A) Clinical expertise, patient satisfaction, cost reduction

B) Best evidence, clinical expertise, patient values

C) Data mining, research protocols, patient safety

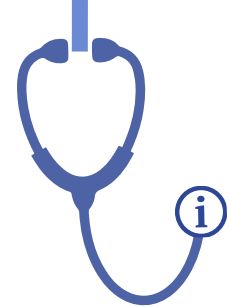
6-Which of the following is NOT a benefit of implementing Medical Informatics in healthcare?

A) Improved healthcare quality

B) Reduction in redundant tests and errors

C) Complete elimination of medical expertise in clinical decision-making

1-C. 2-B. 3-C. 4-B. 5-B. 6-C.





Team Leaders

Shahad Alshehri



Khalid Alghamdi

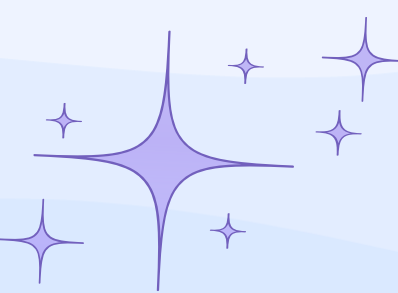


Team Members

Sahar Alfallaj



Mohammed Aljammaz



سبحان الله، الحمد لله، لا إله إلا الله، الله أكبر، لا
حول ولا قوة إلا بالله

